Assignment 3: Distributed Batch Processing Using Apache Spark

CS328 - Distributed and Cloud Computing

Deadline: 23:59, Dec 14, 2025

1 Requirements

You are provided with the dataset (parking_data_sz.zip) containing information on parking lot utilisation in Shenzhen. The dataset is in .csv format, with the entries (lines) storing information about the parking location and parking duration of a vehicle. The following table describes the data stored for each vehicle (columns):

Name	**Description**	**Example**
in_time	Time vehicle entered the parking lot	"2018-09-01 10:10:00"
out_time	Time vehicle left the parking lot	"2018-09-01 12:00:00"
berthage	Unique ID of the lot used by the vehicle	"201091"
section	Section of the lot used by the vehicle	" 荔园路 (蛇口西段)"
admin_region	District of the lot used by the vehicle	" 南山区"

You are required to perform five data analysis tasks on this dataset using *Apache Spark's* Python API, *PySpark*, and output the results in five separate .csv files. Specifically:

1. Output the total number of *berthages* for each *section*. The output file should have the columns **section** and **count**. (10 points)

2. Output all unique *berthages* for each *section*. The output file should have the columns **berthage** and **section**. (20 points)

3. Output the average parking time for each *section*. The output file should have the columns **section** and **avg_parking_time**. The average parking time should be an integer representing seconds (i.e., cast the value to Python `int`). (20 points)

4. Output the average parking time for each *berthage*, sorted in descending order. The output file should have the columns **berthage** and **avg_parking_time**. The average parking time should be an integer representing seconds (i.e., cast the value to Python `int`). (20 points)

5. For each *section*, output the total number of *berthages* in use and the percentage of in-use *berthages* in that *section*, in a one-hour interval (e.g. during 09:00:00-10:00:00). "In use" means there is at least one vehicle parked in that *berthage* in that interval for ANY duration. The time intervals should begin at the start of the first hour that appears in the dataset (for example, if the first entry is at 12:23, the first interval should start from 12:00). The output file should have the columns **`start_time`**, **`end_time`**, **`section`**, **`count`** and **`percentage`**. The percentage value should be rounded to one decimal place (e.g. 67.8%). The data format of **`start_time`** and **`end_time`** should match the one in the original dataset "`YYYY-MM-DD HH:MM:SS`" (20 points)

   - **Sub-task:** Select **three** sections from the results and plot a figure with time as the X axis and percentage as the Y axis. Analyse the results and state any interesting trends that you identify in the report. (10 points)

For each task explain the implementation logic and the *PySpark* calls you used for your computation.

For the above, you can use the Standalone version of PySpark in combination with your local file system.

1.1 Implementation notes

- This assignment does not impose any requirement on the data representation used (RDD or DataFrames); you can choose whichever is more convenient for the task. When working with DataFrames, you can use both Spark SQL and work with the DataFrame methods depending on your preference.
- Both .py scripts and .ipynb notebooks are allowed. Use whichever you prefer. If you use comments to explain your code, write them in English.

2 Limitations

- Using pure Python and Python libraries, excluding PySpark, for data processing is not allowed. **ALL** data processing **MUST** be done using the PySpark API. Sparks' "user-defined functions (UDF)" are allowed. Pure Python and relevant libraries can be used to write the `.csv` files with the results and for plotting.
- There is no 'hard' limitation on how many transformations and actions you can use to complete the tasks. However, you should aim for the simplest solution. Highly convoluted solutions that could be easily avoided by using a transformation already provided by Spark can result in a loss of marks. It is recommended to explore Spark's documentation to find the best transformations for each task.

3 Hints

1. Filter out invalid data (e.g., `out_time` $\leq$ `in_time`) in advance.
2. Consider if converting the data (`in_time`, `out_time`) to (`in_time`, `parking_time_length`) can simplify some tasks.
3. Visit:
   - https://spark.apache.org/docs/latest/api/python/reference/pyspark.html
   - https://spark.apache.org/docs/latest/api/python/reference/pyspark.sql/index.html

   to find useful PySpark RRD and DataFrame transformations and actions.

4 What to Submit

1. Source code: `.py` scipt(s) OR `.ipynb` notebook(s).
2. A report in **PDF** format.
3. Five separate `.csv` files containing the results of the five requirements. The names of the files should be: `r1.csv, r2.csv, r3.csv, r4.csv, r5.csv`. Please pay close attention to ensure the CSVs are named and formatted properl.

Pack all files into `SID_NAME_A3.zip`, where `SID` is your student ID and `NAME` is your name (e.g., 11710106_ 张三 _A3.zip).